

- 53. Heart Pacemaker** A heart pacemaker consists of a switch, a battery of constant voltage E_0 , a capacitor with constant capacitance C , and the heart as a resistor with constant resistance R . When the switch is closed, the capacitor charges; when the switch is open, the capacitor discharges, sending an electrical stimulus to the heart. During the time the heart is being stimulated, the voltage E across the heart satisfies the linear differential equation

$$\frac{dE}{dt} = -\frac{1}{RC} E.$$

Solve the DE, subject to $E(4) = E_0$.

Soln:

$$\frac{dE}{dt} = -\frac{1}{RC} E$$

$$\Rightarrow \int \frac{dE}{E} = -\frac{1}{RC} \int dt$$

$$\Rightarrow \ln E = -\frac{t}{RC} + C_1$$

$$\Rightarrow E = e^{C_1} \cdot e^{-\frac{t}{RC}}$$

$$\Rightarrow E = k e^{-\frac{t}{RC}} \quad \left[\text{Assuming, } k = e^{C_1} \right]$$

$$\text{if, } E(4) = E_0$$

$$\therefore E_0 = k e^{-\frac{4}{Rc}}$$

$$\Rightarrow k = E_0 e^{\frac{4}{Rc}}$$

$$\therefore E = k e^{-\frac{t}{Rc}}$$

$$= E_0 e^{\frac{4}{Rc}} \cdot e^{-\frac{t}{Rc}}$$

$$\Rightarrow E = E_0 e^{\frac{4-t}{Rc}}$$